

General Guidelines for Regulating Data Center Noise

A policy framework for addressing low-frequency industrial noise from data centers in municipal zoning and noise ordinances. Prepared August 2026, amended September 2026 by Kasey Fowler-Finn, Ph.D. Professor of Biology, Saint Louis University. Sound researcher since 2001.

This document reflects the author's independent analysis and does not represent an official position of her institution or funding agencies.

Purpose

Data centers are among the fastest-growing land uses in American cities, and most existing municipal noise ordinances were not built to regulate them. Standard noise codes are typically designed around traffic, construction, and other broadband, higher-frequency sources — not the continuous, low-frequency hum produced by industrial-scale cooling equipment operating 24 hours a day, 7 days a week.

This document summarizes the physical and health-science evidence explaining why data center noise is a distinct regulatory challenge, identifies the gaps common to many existing and proposed ordinances, and offers a set of policy elements that any jurisdiction can adapt when writing or amending data center zoning and noise regulations.

Noise complaints from data center neighbors aren't rare or isolated — they're close to the default experience wherever a large facility operates near housing. Community friction over data center noise has been documented in dozens of states, from Virginia and Georgia to Texas, Ohio, Wisconsin, and Mississippi, with residents in unrelated communities consistently describing the same thing: a persistent hum that doesn't register as a problem on a standard sound reading but that they can hear, and are bothered by, every day. That same complaint surfacing independently across communities with no coordination between them is itself evidence of a structural measurement problem, not a handful of unusually sensitive residents.

That pattern escalates well beyond informal complaints, too. Of the class-action lawsuits filed against data center operators in 2026, the clear majority — five of six, as of this writing — were specifically about noise, not water use, land use, or any other complaint category, and four remain active. That holds across very different jurisdictions (Mississippi, Wisconsin, New Jersey, Virginia among them), which suggests the underlying problem is the noise itself, not any single city's particular ordinance.

The clearest illustration of the problem: a facility's overall noise reading can be completely unremarkable — comfortably within any dBA or dBC limit — while still producing one loud, narrow-band hum that residents notice immediately, because a single decibel number averages a distinct tone in with everything quieter around it. “Is this room loud overall?” and “is there a particular type of noise that is loud?” are different questions, and only the second one predicts resident complaints. This distinction — not any unsettled question about health effects — is the most defensible reason BB49's noise provisions need amending, since it does not depend on resolving any contested science; it rests only on what a single-number measurement can and cannot show. Section 4 explains the mechanics.

1. Low-Frequency Noise Behaves Differently

In general, low-frequency noise from data centers is a special policy concern because it (i) travels farther and is substantially harder to mitigate than ordinary industrial or traffic noise, and (ii) is systematically undercounted by the dBA-based metrics most existing ordinances use to measure compliance. This means a facility can be

compliant on paper while still delivering audible — or, for very low frequencies, inaudible but still physiologically relevant — daytime and nighttime noise exposure linked to numerous health concerns.

Sound attenuates — becomes quieter — as it travels from a source for two distinct reasons. The first is geometric spreading: sound waves lose about 6 dB every time the distance from the source doubles, a purely geometric effect from the sound energy spreading over an ever-larger area, and it applies equally at every frequency (from a whistle to a deep bass) and typically accounts for 30 dB or more of total loss by 3,000 feet. The second is absorption: sound energy dissipating into the air, ground, and any objects in its path, and it increases sharply with frequency — high frequencies lose far more to the air itself than low frequencies do. The same wavelength-dependence shows up again, at much larger scale, wherever sound meets a barrier: a wall or berm only blocks sound by diffraction, bending it around an edge, and it can only do that well when it's large compared to the wavelength trying to get past it. Open air and solid barriers fail low frequencies for the same underlying reason — nothing in ordinary noise control, natural or built, is sized for wavelengths that long. [1]

Consequently, low frequencies propagate over much greater distances than higher frequencies, and they pass through walls and windows with little loss, since ordinary building materials are, for practical purposes, ineffective at attenuating them. Low-frequency sound can also generate secondary noise by inducing vibrations and rattles in nearby structures. [1,2]

Typical building enclosures like solid walls fail for a second reason: mass. Wall performance follows the “mass law” — transmission loss drops roughly 6 dB every time frequency is halved, so a wall built to stop ordinary traffic or industrial noise performs substantially worse at the low frequencies data center cooling equipment produces. Recovering that loss requires roughly doubling the wall's mass for every 6 dB needed, not simply adding thickness of a lighter material.

Absorptive lining fails for a third reason: effectively absorbing a given frequency requires material roughly a quarter-wavelength thick. At 1,000 Hz that's about four inches — easy. At 63 Hz it's roughly 4.5 feet; at 20 Hz, roughly 14 feet — not a remotely practical thickness for a building enclosure. [1]

Practical implication: setbacks and enclosure requirements calibrated to ordinary industrial or traffic noise — which is well attenuated by both distance and standard enclosures — will systematically under-protect residents from the specific noise signature of data center cooling equipment: a continuous, low-frequency, tonal hum that conventional mitigation was never designed to stop.

2. Propagation Distances

- While the most severe impacts of noise from data centers and similar types of low-frequency producing equipment are concentrated within 3,000 feet, noise from data centers has been reported as audible up to 2-2.5 miles from facilities [4,5]. There are concrete reasons for this radius that can be explained by the physics of sound that help explain why complaints related to sound regularly occur at distances beyond those that free-spreading calculations alone would predict.
- ISO 9613-1 (international standard) only tabulates absorption coefficients down to 50 Hz, because below 50 Hz is considered too low to matter for standard noise engineering. Standard noise engineering is built on typical sources of environmental noise (traffic, block parties, etc.) that generate noise in higher frequency ranges than those produced by data centers. Noises below 50Hz are problematic from a human health perspective even though they are not captured by traditional dBA metrics.
- Temperature inversions are common at night (during the day the ground is often hotter than the air, but this reverses at night). Because temperature affects the speed at which sound travels, these

inversions actually bend sound back towards the ground instead of it simply traveling upward and outward from the source. From a practical point of view, this leads to sound regularly traveling to distances that free-spreading calculations alone would predict to be inaudible. This is why complaints often originate at further distances than “expected.” It also means that a facility could pass a noise test at 2pm but not 3am — and it is the cleanest justification available for requiring night-specific, multi-condition testing rather than a single fair-weather reading (see Recommendation 4). [12]

- Wind patterns can also affect sound transmission, with areas downwind experiencing greater noise levels at further distances than expected. Jurisdictions should assess local prevailing wind patterns when siting facilities and setting setback requirements, since downwind residential areas can experience meaningfully elevated noise exposure relative to upwind areas at the same distance. [12]
- Many municipal setback requirements for data centers, where they exist at all, fall in the range of roughly 100–600 feet from residentially zoned parcels, schools, and parks. These distances are typically well within the range where low-frequency noise impacts have been documented and measured in the literature [3,5]. This is because these setbacks are generally written for ordinary or broadband industrial noise profiles and are, quite simply, not far enough for the noise signature data centers actually produce.

TAKE HOME: jurisdictions cannot set a “safe” setback distance for low-frequency noise using the same intuitions and calculations used for “ordinary” noise.

3. Health Evidence

The World Health Organization has conducted several longitudinal studies and systematic reviews showing that environmental noise exposure increases health risk in a dose-dependent way [7,8,9,10]. Sleep disruption, annoyance, hearing loss, cardiovascular disease, and impaired cognition in children are all known effects of environmental noise [7,8,9,15,16]. The thresholds (≥ 40 dB at night ≥ 55 dB during the day) [8,11] are levels above which adverse health effects become a population-level concern [11]. At data center property lines, noise routinely measures 55–65 dBA [4], a range meeting or exceeding WHO nighttime adverse-effects threshold every single night these facilities operate. Importantly, these figures are based on dBA measurements, which drastically under-measure low frequencies (see more on this below).

Studies specifically focused on the effects of low-frequency noise are less abundant than those for broad band noise. This is because of the low sensitivity of standard measurement to this frequency range rather than to an absence of effect. In fact, reviews assessing multiple studies focused on low-frequency noise confirm sleep disruption, cardiovascular effects, and annoyance that parallel the effects of exposure to broadband noise [13,14].

TAKE HOME: Low-frequency noise has similar negative effects on human health as higher-frequency noise. Because it evades standard dBA measurement and resists conventional mitigation, it can deliver the sustained exposure levels the WHO evidence links to cardiovascular, sleep, and cognitive harm while still registering as “compliant” under a dBA/DBC-only standard.

4. Common Gaps in Existing and Proposed Ordinances

Those municipal noise ordinances that actually address data centers typically use only dBA measurements, and occasionally DBC measurements. A typical baseline noise level provision defines noise as a dBA (A-weighted) and/or DBC (C-weighted) measurement averaged over some fixed period (commonly 30–60 minutes), broken into daytime and nighttime windows. After construction is complete and a facility is running, compliance standards typically cap noise levels at a fixed number of decibels (often 3–5 dB) above that baseline.

Neither dBA nor dBC measurements are designed to capture low frequency sound with equal fidelity across the range. dBA is specifically designed to de-emphasize it — down roughly 50 dB at 20 Hz — which is the far more significant gap of the two. C-weighting is flatter and stays within about a decibel of accurate down to roughly 30–63 Hz, with its deviation only becoming acoustically meaningful below approximately 30 Hz. G-weighting (ISO 7196) is sometimes proposed as a fix for this gap, but it is not a good fit here: it is purpose-built for true infrasound in the 1–20 Hz range and rolls off at 24 dB per octave above 20 Hz — roughly 40 dB down by 63 Hz — with a perception threshold near 100 dB(G). A G-weighting requirement would hand a facility a compliance metric that is nearly impossible to fail and that goes nearly deaf to exactly the 30–125 Hz range where data center cooling equipment noise concentrates. The better tool for this range is not a different weighting curve at all, but a different kind of measurement entirely: octave- or third-octave-band analysis, which does not break down sound into a weighted single number in the first place.

A regular noise meter gives a single number (e.g., 69 dB) that represents all the sound happening across all frequencies at once, whatever weighting curve is applied. This approach becomes problematic if there is a disruptive set of frequencies that are particularly loud, because the total dB across all frequencies hides particularly disruptive noise in narrow frequency bands, as they get blended in with the rest. Octave-band measurements fix this by splitting sound up by pitch/frequency bands before measuring it. Instead of one number for all frequencies, you get a separate reading for each specific frequency range: how loud is it between 16–31 Hz, how loud between 31–63 Hz, how loud between 63–125 Hz, and so on, each range roughly doubling in pitch from the last. Third-octave bands do the same thing but slice each of those ranges into three smaller pieces, for even more precision.

The result is that instead of assessing how loud the noise is overall, you can ask if specific frequency ranges are particularly loud, which is actually what matters for complaints because a facility's overall dBA or dBC number can look completely unremarkable because the one loud, droning frequency band is averaged in with several quieter ones. If you run an octave-band analysis on the same noise, that same droning hum shows up as an obvious spike in one specific band. The exact thing residents are actually hearing and complaining about is made visible instead of hidden inside an average, because a distinct tone or hum is far more noticeable and annoying than the same total energy spread evenly across all frequencies.

Additional gaps commonly found in existing and proposed ordinances:

- Baseline noise levels are frequently self-measured by the applicant pre-application, with no requirement for third-party verification — creating an incentive to establish an inflated baseline against which future compliance is measured.
- Post-construction compliance is often checked via a single annual third-party report, sometimes taken only during a narrow seasonal window — missing seasonal variation in cooling load and any nighttime-specific degradation outside that window.
- A flat allowance above baseline (commonly 3–5 dB) is a single number applied to a source type (continuous low-frequency hum against a quiet nighttime baseline) where small increases are disproportionately perceptible, and where the underlying measurement itself is acknowledged industry-wide to be difficult with standard equipment — raising the risk that a real, perceptible increase falls within what gets treated as measurement uncertainty [1,14].
- Setback distances are frequently not tied to any frequency-specific propagation modeling and are much shorter than distances at which independent measurement (the wind-turbine literature, the closest analog to data centers) has detected a measurable low-frequency contribution from an operating facility [3].

5. Recommended Policy Elements

Policy Element 1: Require a third-octave- or octave-band measurement with defined compliance thresholds as a mandatory component of every required noise measurement — not merely dBA/dBC.

dBA weighting alone is inadequate to regulate the frequency range data center cooling equipment actually produces: it is specifically designed to de-emphasize sound below 500 Hz, down roughly 35 dB at 40 Hz. dBC is a meaningful improvement and should be retained, not replaced — it captures lower frequencies better — but neither dBA nor dBC, alone or together, can reveal a narrow tonal spike hidden inside a broadband average, the exact type of noise that residents complain about. Ordinances should require, in addition to the existing dBA/dBC baseline: (a) third-octave band sound pressure levels from 20–200 Hz (per ISO 9613- 1/9613-2 methodology), so that any tonal or narrow-band component of the facility's noise is independently identifiable rather than averaged away into a single broadband number — the 20 Hz floor reflects the practical limit of reliable outdoor third-octave measurement; and (b) a defined compliance ceiling — for example, noise shall not exceed 5 dB above baseline in any individual third-octave band — so that a facility cannot pass an overall dBC allowance by averaging a single loud low-frequency band against several quieter ones. Ordinances should also state explicitly that compliance with a dBA/dBC-based standard alone does not constitute compliance with this Chapter absent also meeting the third-octave-band criteria above, and this measurement requirement should apply at every stage of a facility's lifecycle: pre-application baseline, application-stage projections, annual compliance reporting, and any complaint-triggered re-measurement.

Policy Element 2: Require independent, City-selected or -approved, applicant-funded acoustical assessment both pre- and post-construction.

Both the pre-application baseline noise level and any application-stage noise projections should be measured or verified by an independent acoustical engineer selected or approved by the municipality — rather than supplied solely by the applicant — with the cost borne by the applicant. The same requirement should extend to post-construction compliance verification, so that any required annual or periodic noise report is performed by an engineer selected by the City (not simply an engineer “acceptable to” a City official, which leaves selection ambiguous) and paid for by the applicant. This structure — City selection, applicant funding — removes the incentive problem inherent in a self-selected or self-measured baseline and ensures a consistent, independent standard applies from initial application through ongoing operation.

Policy Element 3: Set a stricter, absolute nighttime noise ceiling, independent of baseline.

A uniform day/night allowance above baseline is not adequately protective for the hours that matter most. WHO's $L_{\text{night}} \geq 50$ dB threshold for adverse health effects applies specifically to nighttime exposure, and the sleep-disturbance and cardiovascular evidence above is drawn overwhelmingly from nighttime exposure conditions. A flat day/night allowance also ignores the atmospheric effect described in Section 2 that works specifically against residents at night: temperature inversions commonly refract sound back toward the ground during calm, clear nighttime conditions, meaning the same facility can be measurably louder at 3 a.m. than a daytime compliance test would predict. Ordinances should instead adopt: a stricter nighttime-specific allowance above baseline (e.g., 3 dB rather than 5 dB); an absolute nighttime ceiling at the property line (to be established in consultation with the jurisdiction's acoustical engineer, informed by the WHO night threshold), regardless of baseline, so a facility with an already-elevated baseline cannot use that baseline to justify nighttime levels above WHO's adverse-effects threshold; extension of both the delta allowance and the absolute ceiling to the third-octave-band metrics in Policy Element 1 — an absolute per-band floor as well as a relative one — so a facility cannot satisfy the nighttime standard on a broadband basis while still producing an audible low-frequency tone during peak sleep-disturbance hours; and a requirement that nighttime compliance measurement occur across multiple nights and seasons, rather than a single reading, to capture inversion and calm-wind conditions rather than allowing one fair-weather night to establish year-round compliance. Daytime allowances may reasonably remain more permissive, since the strongest dose-response evidence centers specifically on the nighttime window.

Policy Element 4: Require continuous or seasonally-representative monitoring, not a single annual snapshot.

Ordinances should require either continuous property-line monitoring (increasingly low-cost and standard for this purpose) or, at minimum, quarterly measurement windows covering all seasons, with results reported to the relevant health or environmental authority on a rolling basis and made available to affected residents.

Policy Element 5: Set setback distances based on low-frequency-specific propagation data, not generic industrial-noise assumptions.

Setback distances should not be set using intuitions calibrated to ordinary broadband industrial or traffic noise. Health Canada's Wind Turbine Noise and Health Study — one of the most rigorous independent, nonindustry-funded dataset available for a comparable continuous low-frequency source — found measurable low-frequency and infrasound contributions persisting to distances of multiple kilometers, decaying only gradually (roughly 3 dB per doubling of distance beyond 1 km). Advocacy and journalistic reporting independently documents data center noise as audible up to two miles from a facility, with the most severe impacts concentrated within roughly 3,000 feet. Given that low-frequency noise is not meaningfully attenuated by distance the way ordinary noise is, jurisdictions should set setback distances well above what generic industrial-noise assumptions would suggest — on the order of 1,500 feet or more for the largest facility classifications, scaled proportionally for smaller classifications — unless a facility's third-party pre-application acoustic assessment demonstrates it will not produce a measurable low-frequency increase at a shorter proposed distance. Any mandated periodic policy review should be genuinely mandatory, with a specific charge to reassess setback adequacy against updated propagation data and accumulated complaint or compliance history, and with findings requiring a governing-body vote on any recommended amendment.

Policy Element 6: Build in a mandatory, applicant-funded, complaint-triggered re-testing mechanism with defined thresholds, deadlines, and enforcement consequences.

A vague “no unabated nuisance violations” standard, without a defined path from a resident complaint to an actual re-test, leaves enforcement entirely discretionary and reactive. Ordinances should instead specify: a trigger (e.g., five or more documented resident complaints from separate households within any 90-day period, logged through a public-facing system, automatically triggering a mandatory re-measurement regardless of whether staff independently judge the complaints credible); a timeline (e.g., a City-selected acoustical engineer completing re-measurement — including G-weighted/octave-band metrics — within 30 days of the trigger, with public results within 10 additional days); funding (the full cost borne by the facility operator, not the City or complainants); consequences for violation (a defined correction period, e.g., 60 days, followed by mandatory re-testing at the operator's expense, with a second confirmed violation within 12 months constituting grounds for permit revocation proceedings); and an anti-gaming provision specifying that private settlement agreements or non-disclosure agreements between an operator and a resident cannot be used to suppress complaints from counting toward the threshold.

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REFERENCE GLOSSARY

Acoustics & Noise Policy: Glossary of Terms

Plain-language definitions of the acoustics terms used in data center noise policy discussions.

Prepared by Kasey Fowler-Finn, Ph.D.

How Sound Travels and Fades

Attenuation: The general term for sound getting quieter as it travels. Attenuation happens through two distinct mechanisms — geometric spreading and absorption — described below. Both happen at once, but their relative importance changes a lot depending on the frequency (pitch) of the sound.

Geometric spreading: Sound loses about 6 decibels (dB) every time the distance from the source doubles, simply because the sound energy is spreading out over an ever-larger area — the same way a flashlight beam gets dimmer and covers more area the farther it travels. This happens equally at every frequency and can't be blocked or engineered around; it's pure geometry.

Absorption: Sound energy converting into a tiny amount of heat as it passes through air, the ground, or a solid object like a wall or a stand of trees. Unlike geometric spreading, absorption is strongly affected by frequency — high-pitched sounds are absorbed quickly, while low-pitched sounds pass through with much less loss. This is the mechanism that walls, berms, and acoustic foam are designed to use.

Temperature inversion: An atmospheric condition, common on calm nights, where cooler air sits near the ground beneath warmer air above — the reverse of the typical daytime pattern. Because sound travels faster in warmer air, an inversion bends sound waves back down toward the ground instead of letting them spread upward and away, allowing noise to travel farther and stay louder at night than daytime measurements would predict.

Frequency and Pitch

Frequency: How many times a sound wave vibrates per second, measured in Hertz (Hz). Higher frequency = higher pitch. A deep rumble might be 20–50 Hz; a human voice is roughly 100–250 Hz at its fundamental pitch; a smoke detector beep might be 3,000+ Hz.

Low-frequency noise: Generally, sound below roughly 200–250 Hz. This is the range dominated by industrial equipment like data center cooling fans and chillers — a continuous, deep hum rather than a sharp or hissing sound.

Infrasound: Sound below about 20 Hz — technically below the typical threshold of human hearing, though very intense infrasound can still be felt or physiologically registered even when it isn't consciously heard.

Wavelength: The physical length of one full cycle of a sound wave, calculated as speed of sound ÷ frequency. Low-frequency sounds have long wavelengths (a 20 Hz wave is about 17 meters, or 56 feet, long); high-frequency sounds have short wavelengths (a 1,000 Hz wave is about 1 foot long). Wavelength is the reason low frequencies are so hard to block — see “quarter-wavelength rule” below.

How Noise Gets Mitigated — and Why It Often Fails at Low Frequencies

Quarter-wavelength rule: A rule of thumb in acoustics: absorptive material (foam, fiberglass, mineral wool) needs to be roughly a quarter as thick as the sound's wavelength to meaningfully absorb it. For a 1,000 Hz sound, that's about 4 inches of material — easy. For a 20 Hz sound, that's roughly 14 feet of material — not remotely practical for a building enclosure.

Absorption (as a mitigation strategy): Lining a wall, ceiling, or enclosure with soft, porous material to soak up sound energy as it passes through. Effective for high-pitched sound; effectively useless for low-frequency sound because of the quarter-wavelength problem above.

Mass-based blocking (transmission loss): How a solid, dense wall keeps sound from passing through it — the sound wave has to physically move the wall to get through, and a heavier wall is harder to move. This is a completely different mechanism from absorption; it depends on the wall's weight, not its thickness.

Mass law: The acoustic principle describing how well a solid wall blocks sound based on its mass. Under the mass law, a wall's blocking performance (transmission loss) drops by roughly 6 dB every time the frequency is halved — meaning a wall that blocks 1,000 Hz well will perform substantially worse at 500 Hz, worse still at 250 Hz, and so on. Recovering that lost performance requires roughly doubling the wall's mass for every 6 dB needed — usually done with denser materials (concrete, mass-loaded vinyl) rather than a thicker but lighter wall.

Setback: The minimum distance a piece of noise-generating equipment or infrastructure must be located from a property line, residence, school, or other protected use. Setbacks are a zoning tool meant to use distance itself as a form of noise mitigation — but because low-frequency sound is barely reduced by geometric spreading (see above), setbacks calibrated to ordinary noise sources are often far too short to meaningfully protect against low-frequency sources.

How Noise Gets Measured

Decibel (dB): The standard unit for measuring sound pressure level. The decibel scale is logarithmic, not linear — a 10 dB increase represents roughly a doubling of perceived loudness, and a small numeric change can represent a large real-world difference in sound energy.

dBA (A-weighted decibels): The most common noise-measurement standard, designed to approximate how the human ear perceives loudness across frequencies. **dBA deliberately downplays low frequencies — by around 50 dB at 20 Hz** — because the human ear is naturally less sensitive to them. It's built around frequencies typical of traffic and voices, not deep industrial hums. This makes dBA a poor tool for capturing low-frequency industrial noise, even when that noise is a real health concern.

dB(C-weighted decibels): A less common noise-measurement standard that is flatter across frequencies than dBA and captures more low-pitched sound, but it still rolls off increasingly below about 63 Hz — right where data center cooling noise concerns concentrate.

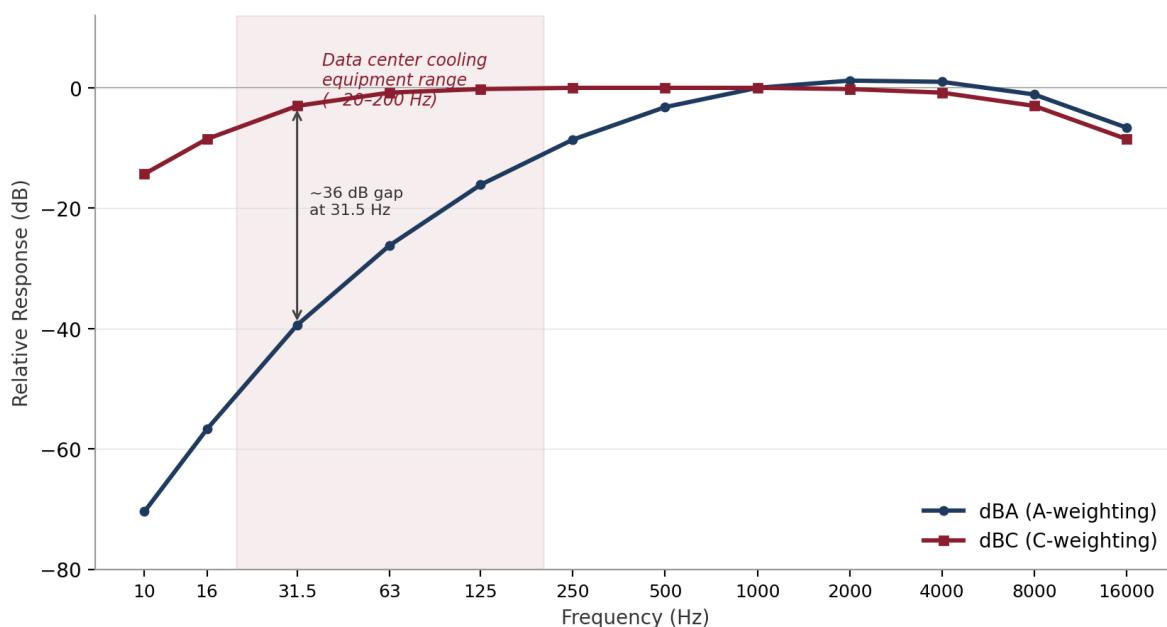
- **Both dBA and dB(C)** produce a single combined number for all frequencies at once. A loud, narrow band of low-frequency noise can be averaged in with quieter readings elsewhere in the spectrum and disappear into an unremarkable overall reading — even though residents nearby can clearly hear it as a distinct hum or tone.

G-weighting (ISO 7196): An international measurement standard purpose-built for evaluating infrasound and very low-frequency sound, unlike dBA or dB(C), which were designed around the frequencies of ordinary environmental noise like traffic and voices.

• **Octave-band and third-octave-band measurement** solves this by reporting noise levels separately across specific frequency ranges rather than blending everything into one number — the difference between asking “how loud is this room overall?” and “is there a particular type of sound that's loud?” The second question is the one that predicts resident complaints, since a distinct tone is far more noticeable than the same energy spread evenly across the spectrum.

Baseline noise level: A measurement of ambient (background) noise at a site, typically taken before a facility is built, used as the reference point against which future noise from the facility is compared for compliance purposes.

How dBA and dBC Weighting Discount Low-Frequency Sound



Standard relative-response values per IEC 61672-1. Both curves are normalized to 0 dB at 1,000 Hz. A more negative value means the weighting suppresses more of that frequency's true loudness.

The gap between the two curves in the shaded region shows how much low-frequency energy dBA discounts relative to dBC — and both still miss the octave-band-level detail that would reveal a narrow, tonal spike within that range.